

Minimum Requirements

To purchase and deliver one (1) large-area 2D X-ray flat panel detector suitable for laboratory cone-beam CT (CBCT) and fluoroscopy imaging to support human-sized phantom development/testing and validation of imaging evaluation methods. Brand name or equal products may be offered, provided all minimum requirements below are met or exceeded.

A) Geometry / sampling

1. Active imaging area (minimum): $\geq 42.7 \text{ cm} \times 42.7 \text{ cm}$.
2. Pixel pitch (maximum): $\leq 170 \text{ }\mu\text{m}$.

B) Image quality (minimum, measured under comparable conditions such as RQA5 or vendor's equivalent)

4. MTF: $\geq 50\%$ at 1 lp/mm (at full-resolution mode).
5. DQE: $\geq 70\%$ at 0 lp/mm (quantum-limited or vendor equivalent definition).
6. Lag: $\leq 3\%$ (first frame).

C) X-ray operating range and performance

7. Energy range compatibility: 40–150 kVp (or wider).
8. Fill factor: $\geq 60\%$.
9. Maximum entrance dose per frame (typical): $\geq$ capability for 50 $\mu\text{Gy}/\text{frame}$ (or better, i.e., requiring less dose for equivalent image quality).

D) Frame rate / acquisition modes (minimum capabilities)

11. Fluoroscopy mode: $\geq 25 \text{ fps}$ at approximately 3x3 binning (or equivalent), with large FOV support.
12. Fluoro zoom mode (if offered): $\geq 30 \text{ fps}$ at approximately 2x2 binning.
13. CBCT projection mode: $\geq 40 \text{ fps}$ at approximately 4x4 binning.
14. Radiography (full resolution) mode: $\geq 4 \text{ fps}$

E) Environmental, mechanical, cooling, and robustness

15. Cooling: Air cooling (no liquid cooling requirement).
16. Operating temperature limit: 19–49°C (or wider).
17. Relative humidity: 10–90% non-condensing (or wider).

- 18. Atmospheric pressure: 70–106 kPa (or wider).
- 19. Radiation tolerance: ≥ 2000 Gy (active area) or better.
- 20. Weight: ≤ 15 kg (or lighter).

F) Interfaces, delivery, warranty, and support

- 22. Complete operational delivery: Must include all items required for operation (detector + required electronics/interface + cables/adapters + documentation) and shipping/handling to [your delivery address].
- 23. Warranty: Minimum 1-year warranty.
- 24. Technical support: Minimum 1-year technical support (remote support required; optional onsite support may be offered as a separately priced line item).